Music Listening Habits During Recessions

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Research Question

How do signals of poor economic conditions affect music listening habits across genres?

Variables

Play Share by Genre

Total Plays

Last.FM API

Recession Indicator

- Binary Variable 1=Recession, 0=No Recession
- NBER, FRED (USREC)

Unemployment Rate

- Continuous Variable, Unemployed Relative to Labor Force
- BLS, FRED (UNRATE)

PCE Price Index

- Continuous Variable, % change = inflation
- BEA, FRED (PCEPI)

UMich Consumer Sentiment Survey

- Continuous Variable, Index of Consumer Sentiment
- University of Michigan, FRED (UMCSENT)

Last.FM

Music application that pairs with streaming services to track personal listening patterns and recommend new music while also acting as a library for listening data.

API Documentation

Confusing encoding...

Remedied by consulting their forum and attending office hours

... then straightforward

Each function had a description and an endpoint

tag

- tag.getInfo
- tag.getSimilar
- tag.getTopAlbums
- tag.getTopArtists
- tag.getTopTags
- tag.getTopTracks
- tag.getWeeklyChartList

- user.getRecentTracks
- user.getTopAlbums
- user.getTopArtists
- user.getTopTags
- user.getTopTracks
- user.getWeeklyAlbumChart
- user.getWeeklyArtistChart
- user.getWeeklyChartList
- user.getWeeklyTrackChart

artist

- artist.addTags
- artist.getCorrection
- artist.getInfo
- artist.getSimilar
- artist.getTags
- artist.getTopAlbums
- artist.getTopTags
- artist.getTopTracks
- artist.removeTag
- artist.search

API Data Collection

tag.getWeeklyChartList → **user.getWeeklyArtistChart** → **artist.getTopTags**

Collected week data (How many weeks am I looking at?)

Collected artist names for every given week.

Got the top tags for each artist

New Variables Created

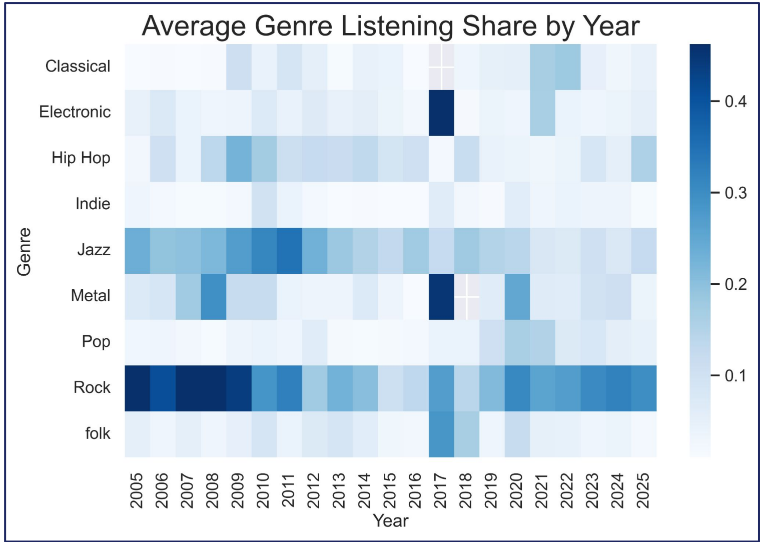
Play Share by Genre

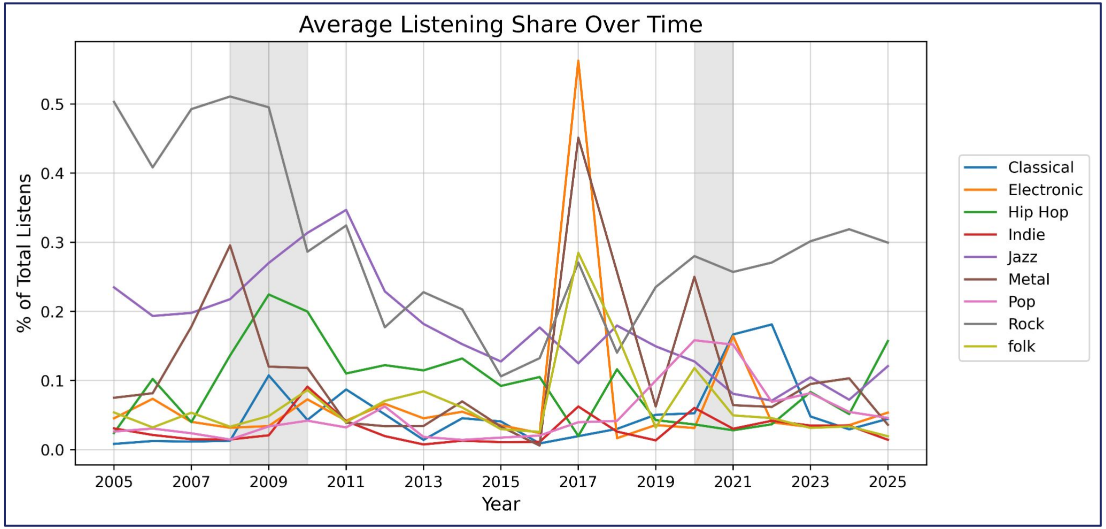
% of total listening shares per week, expressed as decimal

Total Plays

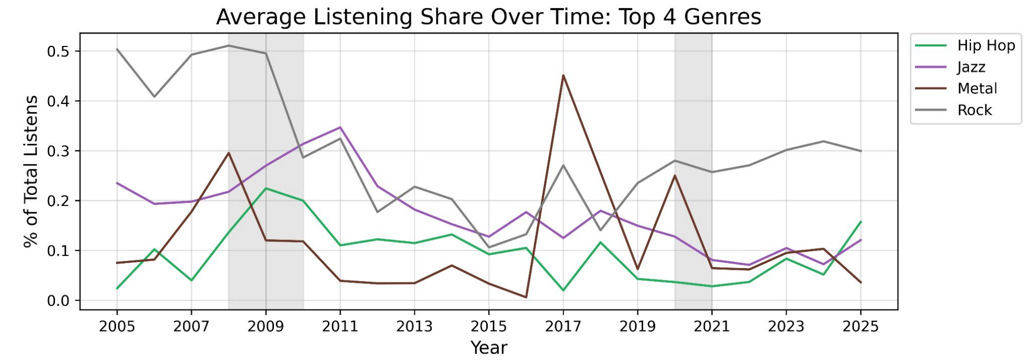
Measures all plays across all artists in a given week of collection

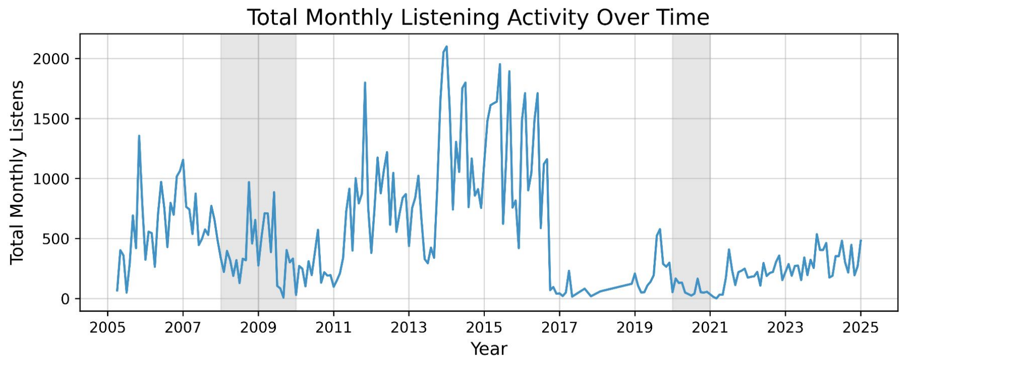
Average Listening Shares by Genre

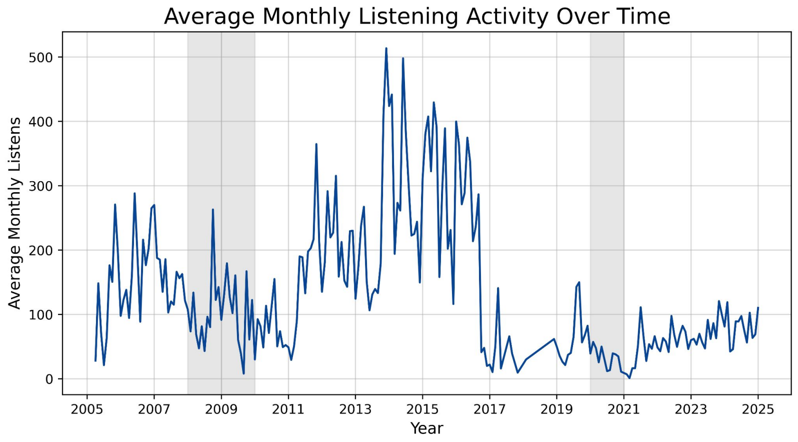




Filtered Graphs







Models

Total Month Plays:

$$Total.Month.Plays = \beta 0 + \beta 1(REC) + \beta 2(UNRATE) + \beta 3\ln(PCE) + \beta 4\ln(UMICH) + \varepsilon$$

Logged Total Month Plays:

$$\ln(Total.Month.Plays) = \beta 0 + \beta 1(REC) + \beta 2(UNRATE) + \beta 3\ln(PCE) + \beta 4\ln(UMICH) + \varepsilon$$

Logged Genre Play Share:

$$\ln(Play.Share_i) = \beta 0 + \beta 1(REC) + \beta 2(UNRATE) + \beta 3\ln(Total.Plays) + \beta 4\ln(PCE) + \beta 5\ln(UMICH) + \varepsilon$$

**i is an individual genre*

Total Plays Regression

Recession's Impact on Total Plays

	Total Plays	
	Total Play Change	% Change
	(1)	(2)
REC	-234.352*	-0.595**
	(120.191)	(0.295)
UNRATE	-1.713	-0.105**
	(17.241)	(0.042)
log(PCE)	-1,067.683***	-3.879***
	(331.641)	(0.815)
log(UMICH)	65.854	-1.574***
	(223.498)	(0.549)
Constant	5,183.800**	31.133***
	(2,296.576)	(5.645)
Observations	232	232
R^2	0.088	0.091
Adjusted R^2	0.072	0.075
Residual Std. Error (df = 227)	447.301	1.100
F Statistic (df = 4; 227)	5.479***	5.675***
Note:	*p<0.1; **p<0.05; ***p<0.01	

Findings

- Recessions <u>**do**</u> cause a decrease in listening (nearly 60%)
- Unemployment has little impact in play change, larger in % change
- Inflation has a very dramatic effect on listening habits
- UMich Survey is only significant when it is logged

Genre Regressions: Top 4

Recession's Impact on Playshare by Genre: Top 4 Genres

	Logged Play Share			
	Rock	Metal	Jazz	Hip Hop
	(1)	(2)	(3)	(4)
REC	-0.094	0.614^{***}	-0.337^{**}	0.235
	(0.145)	(0.223)	(0.161)	(0.301)
UNRATE	-0.139^{***}	-0.150^{***}	0.096^{***}	0.209^{***}
	(0.022)	(0.035)	(0.024)	(0.036)
log(Total.Plays)	-0.181^{***}	-0.720^{***}	-0.111^{***}	-0.257^{***}
	(0.034)	(0.064)	(0.042)	(0.066)
log(PCE)	-4.153^{***}	-3.521^{***}	-3.484^{***}	2.314^{***}
	(0.402)	(0.637)	(0.447)	(0.665)
log(UMICH)	-2.435^{***}	-1.003^{*}	-0.465	1.017^{**}
	(0.283)	(0.517)	(0.326)	(0.475)
Constant	29.673^{***}	21.484^{***}	15.812^{***}	-17.998^{***}
	(2.885)	(4.764)	(3.213)	(4.807)
Observations	805	431	688	505
R^2	0.178	0.315	0.195	0.095
Adjusted R^2	0.173	0.307	0.189	0.086
Residual Std. Error	0.976 (df = 799)	1.141 (df = 425)	0.997 (df = 682)	1.189 (df = 499)
F Statistic	34.620^{***} (df = 5; 799)	39.098^{***} (df = 5; 425)	33.107^{***} (df = 5; 682)	10.446^{***} (df = 5; 499)

Note: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

- Metal and Hip Hop experience <u>**increases**</u> in play share during recessions
- 1 PP increase in Unemployment decreases Rock and Metal, increases Jazz and Hip Hop
- As total plays increase, genres get more diverse
- Inflation again is very dramatic, except for Hip Hop
- UMich decreases all except for Hip Hop

Genre Regressions: Special Cases

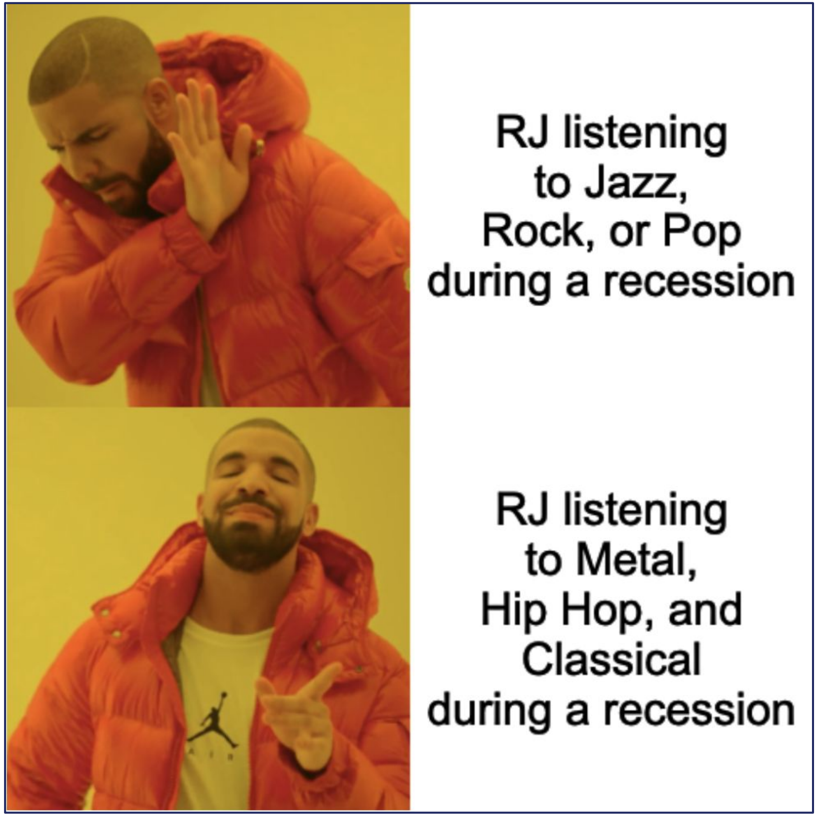
Recession's Impact on Playshare by Genre: Special Cases

	Logged Play Share	
	Pop	Classical
	(1)	(2)
REC	-0.322	0.444
	(0.215)	(0.507)
UNRATE	0.032	0.066
	(0.026)	(0.072)
log(Total.Plays)	-0.768^{***}	-0.689^{***}
	(0.048)	(0.132)
log(PCE)	1.614^{***}	3.727^{**}
	(0.491)	(1.419)
log(UMICH)	0.670^{**}	0.160
	(0.324)	(1.079)
Constant	-10.473^{***}	-18.690^{*}
	(3.456)	(10.343)
Observations	492	106
R^2	0.422	0.351
Adjusted R^2	0.416	0.318
Residual Std. Error	0.894 (df = 486)	1.128 (df = 100)
F Statistic	71.064^{***} (df = 5; 486)	10.813^{***} (df = 5; 100)
Note:		*p<0.1; **p<0.05; ***p<0.01

- Recession Pop does **not** exist for these models, Recession Classical does, however
- Unemployment Rate does not change shares
- Total Plays follows the same trend as other genres
- Inflation increases playshare
- More pop when sentiment increases, possibly more classical

Future Improvements

- **Use actual global data not one person**
- **Take into account song content**
  - Control for mood (happy vs sad rock)
  - Control for keywords
- **Find different method to get genres**
  - User generated tags, not homogenous
- **Use Fixed Effects to control for more popular genres**
  - 'Classic Rock' is going to have a different listen count than PBR&B or Shoegaze, larger more macro genre


RJ listening to Jazz, Rock, or Pop during a recession
RJ listening to Metal, Hip Hop, and Classical during a recession

Thank you!